

FRAA: A Retrieval-Augmented Agent for Explainable Financial Risk Assessment

Abstract. Financial risk assessment requires probabilities that respond to changing behavior and external evidence while remaining transparent enough for human review. Many deployed systems still depend on engineered tabular features and tree ensembles. These models are effective when feature stores are stable, but they incorporate long behavioral histories and newly released market or regulatory information only indirectly. This paper presents the Financial Risk Assessment Agent (FRAA), a retrieval-augmented model that encodes time-aware user behavior, retrieves timestamp-valid financial documents, fuses the retrieved evidence through cross-attention, and returns both a calibrated risk score and a concise justification. On a proprietary 24-month dataset with over 20 million users, 1.2 billion interactions, and 15 data sources, FRAA achieves a Log Loss of 0.5163 and Recall@5 of 0.4185. This corresponds to a 19.6% Log Loss reduction relative to a deployed Random Forest baseline and a 9.4% reduction relative to a non-retrieval Transformer. Ablation and retrieval analyses indicate that timestamp-valid external evidence is the main source of improvement, with five retrieved documents giving the best balance between useful context and retrieval noise. Expert evaluation rates the generated explanations at 4.32/5 for faithfulness, and the tested serving configuration reaches 15.7 ms latency per query.

Keywords: Financial risk assessment · Retrieval-augmented generation · Explainable AI · Risk scoring · Time-aware Transformer

1 Introduction

Financial institutions assess user-level risk from evidence that is heterogeneous, time-varying, and often incomplete at the moment of decision. Such evidence includes transaction histories, account states, macroeconomic indicators, regulatory disclosures, and news sentiment. Existing risk pipelines often separate behavioral modeling, external evidence collection, and explanation generation into different stages, which makes the final decision difficult to update and audit. Reliable risk estimates support credit review and consumer lending decisions [16,38]. They also underpin fraud prevention and compliance workflows [6,31]. In practice, many deployed systems still use carefully engineered tabular features with tree-based models such as Random Forests [7]. Gradient boosting is another standard choice for tabular risk modeling [13,9]. These baselines remain useful, but the central difficulty is not simply fitting a classifier. It is keeping the risk representation synchronized with behavioral changes and newly released external evidence [11,25].

Neural sequence models offer a more flexible way to model user behavior over time, beginning with recurrent architectures such as LSTM and gated recurrent units [17,10]. Transformer encoders further improve long-range sequence modeling [39]. Sequential recommendation work shows their value for evolving user states [22,36]. Similar temporal modeling ideas have been used in credit-card fraud detection [21]. Yet a purely parametric sequence model remains limited in a financial risk setting. It can use market or regulatory information only if that information was encoded during training or manually attached as a feature. It also gives reviewers little direct evidence for why a high-risk case was flagged. Retrieval methods provide a way to condition decisions on external evidence available at inference time. Classical probabilistic retrieval motivates sparse evidence matching [35]. Dense retrieval and RAG extend this idea to neural document representations and generation [23,26]. Fusion-based retrieval models further show that non-parametric evidence can be integrated into downstream prediction [19]. In parallel, local explanation methods have shown how individual predictions can be inspected [34,29]. Broader XAI surveys emphasize transparency as a system-level requirement [14,4]. Calibration studies show that probability estimates require their own evaluation [15,32]. Post-hoc calibration remains relevant when raw classifier scores are not directly reliable probabilities [40]. These observations motivate a risk model that retrieves time-valid financial evidence and links that evidence to both prediction and explanation.

This work develops the **Financial Risk Assessment Agent (FRAA)**, a retrieval-augmented model for explainable financial risk assessment. The word “agent” is used in a constrained operational sense. For each scoring request, FRAA builds a user state, queries a timestamped financial knowledge base, fuses retrieved evidence with behavioral features, estimates a risk probability, and generates a short explanation grounded in the retrieved context and influential feature groups. FRAA is therefore a structured retrieve-fuse-score-explain pipeline rather than an open-ended autonomous system.

The main contributions are:

- We formulate financial risk scoring as a time-constrained retrieval problem and instantiate it with FRAA, which combines temporal user encoding, timestamp-valid evidence retrieval, cross-attention fusion, and explanation generation.
- We evaluate FRAA on a large-scale proprietary multi-source dataset with a strict chronological split, and compare it with deployed tree-based baselines and neural sequence models.
- We report component ablations, feature-group occlusion, retrieval-depth analysis, explanation quality assessment, and latency measurements to examine the contribution and operational cost of the retrieval and explanation components.

2 Related Work

2.1 Financial Risk Modeling and Tabular Baselines

Classical credit-risk systems are built around statistical classifiers, scorecards, and tree ensembles, which remain strong baselines for heterogeneous tabular data [16,38]. Benchmarking studies extend this evidence across credit-scoring datasets and classifier families [3,1]. Later surveys document the breadth of classification methods used in this area [28]. In operational settings, Random Forests and gradient boosting are still widely used because they fit mixed tabular features well [7,13]. XGBoost and large benchmark studies further explain why boosted trees are difficult baselines to displace [9,25]. Credit and fraud datasets also tend to be imbalanced and loss-asymmetric, so ranking quality and calibration matter alongside raw classification accuracy [11,8]. Consumer-credit and fraud studies make the same point in realistic investigation settings [24,5]. Fraud-detection reviews connect these models to monitoring workflows [6,31]. Their strength, however, also defines the boundary of this paradigm: the model state is largely determined by engineered feature stores, while newly emerging market or regulatory evidence must be manually transformed before it can affect a decision.

2.2 Sequential User Modeling

Sequential recommendation and behavior modeling show that recurrent networks can capture evolving user states [17,10]. Transformer encoders extend this capacity with attention-based long-range dependencies [39]. In recommendation, SASRec and BERT4Rec demonstrate the usefulness of self-attention for sequential user modeling [22,36]. Financial risk assessment has a related temporal structure. Recent transactions, account changes, and market signals form irregular sequences whose relevance depends on timing and context. Sequence models have also been applied to credit-card fraud detection, where temporal order is central to identifying abnormal behavior [21]. FRAA follows this line of work but adapts it to calibrated risk scoring, where explanation is part of the target behavior rather than an optional post-processing step.

2.3 Retrieval-Augmented and Explainable AI

Retrieval-augmented methods condition model outputs on non-parametric external evidence. Classical probabilistic retrieval provides the foundation for document matching [35]. Dense passage retrieval and RAG adapt retrieval to neural representations and generation [23,26]. Fusion-in-decoder models further demonstrate that retrieved evidence can be integrated into downstream generation and prediction [19]. Text encoders such as BERT and Sentence-BERT provide practical document representations for semantic matching [12,33]. Approximate nearest-neighbor methods and vector search libraries such as Facebook AI Similarity Search (FAISS) make such retrieval feasible at large scale [30,20]. In regulated domains, however, retrieval alone is not sufficient. Local explanation methods such as LIME and SHAP support instance-level inspection [34,29]. Broader

XAI surveys frame transparency as a modeling and deployment requirement [14,4]. Calibration analyses further show that score reliability must be checked separately from discrimination [15,32]. Post-processing methods remain useful when model scores need to be transformed into calibrated probabilities [40]. Together, these lines of work point to the same requirement: a risk model should expose not only a probability, but also the evidence used to support it.

3 Methodology

3.1 Problem Formulation

For a user u and decision time t , the task is to estimate the probability $\hat{r}_t \in [0, 1]$ that a material adverse event will occur within the next 30 days. The label $y_t \in \{0, 1\}$ indicates whether such an event is observed. The model receives a time-ordered window of behavioral, profile, macroeconomic, and external-document signals available no later than t . This chronological constraint is applied throughout training and testing to avoid future information leakage.

Each training instance is built from an observation window ending at t and a prediction window that immediately follows it. Dynamic events after t are excluded. Slowly changing attributes are read from the latest snapshot available before t . External documents follow the same rule: a document can be retrieved only if its publication or ingestion timestamp is no later than the scoring time. This setup reflects operational risk review, where decisions are made before the adverse event and before later market information becomes available.

3.2 Multi-Source State Construction

FRAA maps the available evidence into three feature families. The dynamic transactional context $\mathbf{d}_t \in \mathbb{R}^{D_d}$ summarizes recent transaction amount, frequency, volatility, merchant category, and event timing. The quasi-static profile context $\mathbf{s}_t \in \mathbb{R}^{D_s}$ contains slowly changing user attributes, including account tenure, product holdings, balance ranges, and risk tier. The market and macro context $\mathbf{m}_t \in \mathbb{R}^{D_m}$ contains time-aligned external indicators available before the scoring time. The external-document branch is motivated by evidence that news and regulatory language can carry risk-relevant signals [37,27].

Let D_d , D_s , and D_m denote the dimensions of the three feature families. These vectors are concatenated and projected into a shared hidden space:

$$\mathbf{x}_t = \mathbf{W}_p[\mathbf{d}_t \parallel \mathbf{s}_t \parallel \mathbf{m}_t] + \mathbf{b}_p, \quad (1)$$

where $\parallel$ denotes concatenation, $\mathbf{W}_p \in \mathbb{R}^{D_h \times (D_d + D_s + D_m)}$ and $\mathbf{b}_p \in \mathbb{R}^{D_h}$ are trainable parameters, and $\mathbf{x}_t \in \mathbb{R}^{D_h}$ is the initial user-state representation.

3.3 Time-Aware Sequence Encoding

Financial behavior is observed at irregular intervals. FRAA therefore injects a time-gap encoding into each event representation. Let $\Delta\tau_i$ denote the elapsed time between consecutive events. The input to layer l of the Transformer encoder is updated as

$$\mathbf{h}_i^{(l)} = \text{TransformerBlock}^{(l)} \left(\mathbf{h}_i^{(l-1)} + \mathbf{p}_{\Delta\tau_i} \right), \quad l = 1, \dots, L, \quad (2)$$

where $\mathbf{p}_{\Delta\tau_i}$ is the time-gap embedding. The final state $\mathbf{h}_t^{(L)}$ summarizes the user’s observed behavior up to time t .

3.4 Retrieval-Augmented Evidence Fusion

To incorporate external knowledge, FRAA converts the encoded user state into a retrieval query:

$$\mathbf{q}_t = f_q(\mathbf{h}_t^{(L)}), \quad (3)$$

where $f_q(\cdot)$ is a lightweight multilayer perceptron that maps the sequence state to the document-embedding space. The financial knowledge base contains approximately 1.2 million timestamped documents, including regulatory filings, market reports, news articles, and macroeconomic commentaries. Documents are embedded offline and indexed with FAISS. At inference time, FRAA retrieves the top- K documents by cosine similarity:

$$\mathcal{D}_t = \text{topK}_{d \in \mathcal{K}_t} \frac{\mathbf{q}_t^\top \mathbf{e}_d}{\|\mathbf{q}_t\| \|\mathbf{e}_d\|}, \quad (4)$$

where $\mathcal{K}_t$ contains only documents available no later than t , and $\mathbf{e}_d$ is the embedding of document d . Document embeddings are produced by a financial-domain text encoder initialized from bidirectional Transformer encoders and sentence-level retrieval models [12,33]. The timestamp restriction prevents future documents from leaking into the prediction.

The retrieved document embeddings are fused with the user state using cross-attention:

$$\tilde{\mathbf{h}}_t = \text{CrossAttn} \left(\mathbf{h}_t^{(L)} \mathbf{W}_Q, \mathbf{D}_t \mathbf{W}_K, \mathbf{D}_t \mathbf{W}_V \right), \quad (5)$$

where $\mathbf{D}_t \in \mathbb{R}^{K \times D_e}$ stacks the embeddings of retrieved documents, D_e is the document-embedding dimension, and $\mathbf{W}_Q$, $\mathbf{W}_K$, and $\mathbf{W}_V$ are trainable projection matrices. The fused representation $\tilde{\mathbf{h}}_t$ is passed to the scoring and explanation heads. Cross-attention is used instead of simple concatenation because the relevance of external evidence depends on the user state. For example, a macroeconomic commentary may matter for users with concentrated sector exposure, but may be weakly relevant for users whose recent transactions do not involve that sector.

3.5 Risk Scoring and Explanation Generation

The risk head maps the fused representation to a calibrated probability:

$$\hat{r}_t = \sigma(\mathbf{w}_r^\top \tilde{\mathbf{h}}_t + b_r). \quad (6)$$

where $\sigma(\cdot)$ is the sigmoid function, $\mathbf{w}_r$ is a trainable risk-head vector, and b_r is a scalar bias. In parallel, a supervised explanation decoder generates a concise textual justification from the retrieved documents and feature-group importance signals. The training objective combines per-instance binary cross-entropy for risk prediction with an explanation-fidelity term:

$$\mathcal{L} = -y_t \log \hat{r}_t - (1 - y_t) \log(1 - \hat{r}_t) + \lambda \mathcal{L}_{\text{exp}}(\hat{e}_t, e_t^{\text{ref}}), \quad (7)$$

where $\hat{e}_t$ is the generated explanation, e_t^{ref} is an expert or rule-supported reference explanation, $\mathcal{L}_{\text{exp}}$ penalizes mismatches with reference evidence and feature-group rationales, and λ balances prediction and explanation supervision. The reference explanation is constructed from analyst-approved risk factors and timestamp-valid evidence snippets, so the explanation loss encourages the decoder to mention evidence that was actually available at the scoring time. In all reported experiments, the Transformer uses 8 attention blocks, hidden dimension 256, and embedding dimension 512. Retrieval depth is $K = 5$ unless otherwise stated.

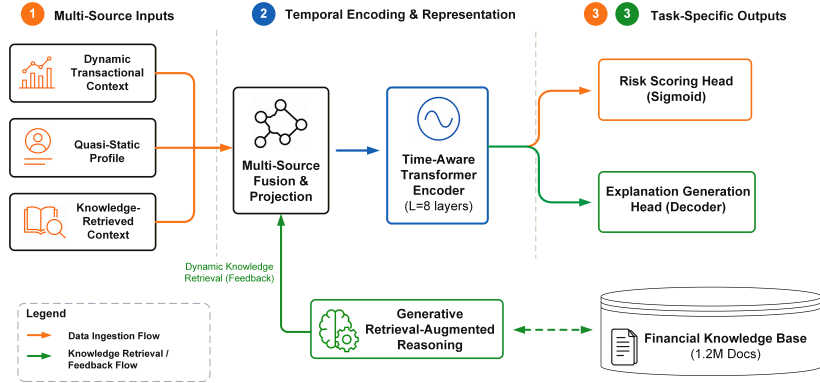


Fig. 1. Overview of FRAA. The model encodes multi-source financial behavior, retrieves timestamp-valid documents from a financial knowledge base, fuses retrieved evidence with the user state, and outputs both a calibrated risk score and an explanation.

Figure 1 summarizes the inference path from multi-source evidence encoding to retrieval, fusion, scoring, and explanation generation.

4 Experiments

4.1 Dataset and Evaluation Protocol

The dataset is a proprietary 24-month multi-source corpus from a financial risk environment. It contains over 20 million users, 1.2 billion interactions, and 15 heterogeneous data sources. Records include transaction events, account status snapshots, product holdings, macroeconomic series, and timestamped news or regulatory signals. The target indicates whether a material adverse event, such as default, severe delinquency, or forced liquidation, occurs within the next 30 days.

We use a strict chronological split. Historical records are used to build look-back features and timestamp-valid knowledge-base entries. The model is trained on a 12-month period, tuned on the following month, and evaluated on the final month. No validation or test-period documents are made available to earlier predictions. We report Log Loss for probability calibration, Recall@1/Recall@5 for risk-review ranking, area under the receiver operating characteristic curve (AUC) for feature-group occlusion, and longest common subsequence ROUGE (ROUGE-L) for explanation overlap with expert-written references.

The evaluation protocol follows the main failure modes of financial risk systems. Log Loss measures whether predicted probabilities are useful for downstream thresholding and risk appetite control. Recall@1 and Recall@5 measure the ranking quality of the highest-risk cases, which are the cases most likely to be reviewed by analysts. The chronological split and timestamp-valid retrieval rule are used for every model variant, including ablations, so that performance differences do not arise from different access to future information.

4.2 Baselines and Training Details

We compare FRAA with a deployed Random Forest (RF), RF with static knowledge-base (KB) features, XGBoost, LSTM, and a non-retrieval Transformer. These baselines cover the main operational choices in the target setting: tabular ensembles, tabular ensembles with manually attached external features, recurrent sequence modeling, and neural sequence modeling without retrieval. We retain the deployed tree-based baseline because credit-scoring studies repeatedly find tree ensembles difficult to displace on heterogeneous tabular data [3,8]. We also include a neural tabular reference point because recent tabular deep learning remains competitive only under careful benchmarking [2]. All baselines use the same chronological split and available feature families. FRAA is implemented in PyTorch and trained with AdamW, learning rate 1×10^{-4} , weight decay 1×10^{-5} , batch size 64, and a maximum budget of 50 epochs. The checkpoint is selected by validation Log Loss, and no test-set information is used for model selection.

Hyperparameters are chosen on the validation month and then fixed for the test month. For retrieval variants, document embeddings are constructed offline and indexed before evaluation. During scoring, the retrieval pool is filtered by

timestamp before nearest-neighbor search. This step is essential because a high-capacity retrieval model could otherwise use documents that were unavailable when the decision should have been made.

4.3 Evaluation Questions

The experiments address three questions. First, does retrieval-augmented evidence change calibrated risk prediction and high-risk ranking relative to tree-based and neural baselines? Second, which components and evidence sources account for the observed gains, especially the dynamic transaction encoder, time-gap encoding, and knowledge retrieval module? Third, are the explanations and latency compatible with analyst-facing risk review? The following sections follow this order, separating the main predictive result from auxiliary analyses.

4.4 Main Results

We first compare the full model with deployed and neural baselines under the same chronological split. Table 1 reports the numerical results, and Fig. 2 visualizes the two primary metrics. The ordering is consistent across calibration and ranking: the full retrieval-augmented model outperforms tree-based baselines, the non-retrieval Transformer, and all ablated FRAA variants.

Table 1. Main test-set performance. Lower Log Loss and higher Recall@5 are better.

Model	Log Loss	Recall@5
RF (deployed baseline)	0.6421	0.3172
RF + KB features	0.6289	0.3291
XGBoost	0.6104	0.3495
LSTM	0.5873	0.3738
Transformer	0.5698	0.3846
FRAA (-Dyn)	0.5642	0.3891
FRAA (-TE)	0.5510	0.3957
FRAA (-KB)	0.5385	0.4023
FRAA (full)	0.5163	0.4185

Relative to the deployed RF baseline, FRAA reduces Log Loss by 19.6% and raises Recall@5 from 0.3172 to 0.4185. Relative to the non-retrieval Transformer, it reduces Log Loss by 9.4% and raises Recall@5 by 8.8%. The ablated variants remain below the full model, indicating that retrieval, temporal encoding, and dynamic features contribute complementary information.

4.5 Ablation and Analysis

Architectural Components The component ablation removes one design choice at a time while keeping the training split, feature availability, and eval-

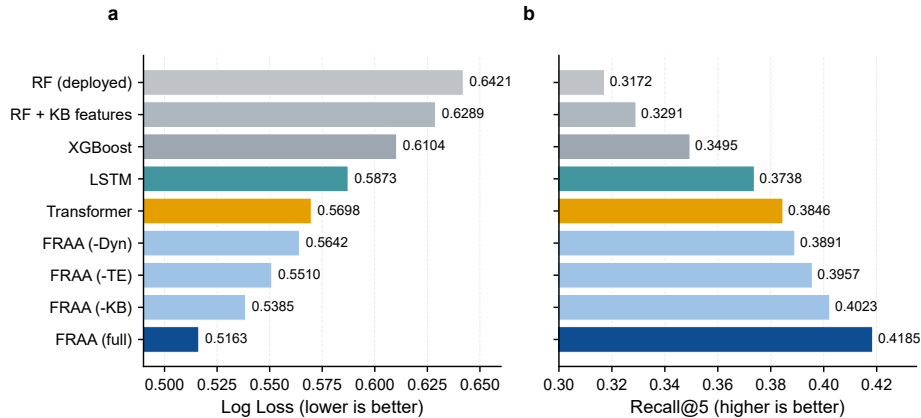


Fig. 2. Main-result comparison. FRAA obtains the lowest Log Loss and highest Recall@5 among all evaluated models.

uation metrics unchanged. This setup separates the effect of retrieval from the effect of stronger sequence modeling.

Table 2. Component ablation on the risk-score prediction task.

Model variant	Log Loss	Recall@5
Transformer	0.5698	0.3846
FRAA (-Dyn)	0.5642	0.3891
FRAA (-TE)	0.5510	0.3957
FRAA (-KB)	0.5385	0.4023
FRAA (full)	0.5163	0.4185

Table 2 isolates the contribution of dynamic transactions (-Dyn), temporal encoding (-TE), and knowledge retrieval (-KB). All three removals degrade the full model, confirming that the gains do not come from a single add-on. Removing the knowledge component increases Log Loss from 0.5163 to 0.5385 and decreases Recall@5 from 0.4185 to 0.4023, supporting the retrieval-augmented design. Removing temporal encoding also reduces performance, which is consistent with the irregular timing of financial events. The -Dyn variant remains stronger than the non-retrieval Transformer. External evidence and temporal encoding therefore compensate for some missing dynamic transaction information, but they do not replace it.

Feature Groups and Retrieval Depth We next examine the gain from two complementary views: feature-family occlusion and retrieval depth. Figure 3

summarizes both analyses using the same values as the feature-family occlusion and retrieval-depth experiments. Knowledge-retrieved context produces the largest AUC drop, followed by transaction dynamics. Static profile variables still contribute, but they are less influential than time-sensitive behavioral and external-evidence signals. Retrieval depth also has a clear optimum in this setting: one or three documents provide insufficient external context, whereas seven or ten documents slightly degrade both metrics. We therefore use $K = 5$ in the main experiments. This pattern is expected in retrieval-augmented risk scoring: too few documents leave the model with limited external context, whereas too many documents introduce loosely related evidence that weakens the fused representation.

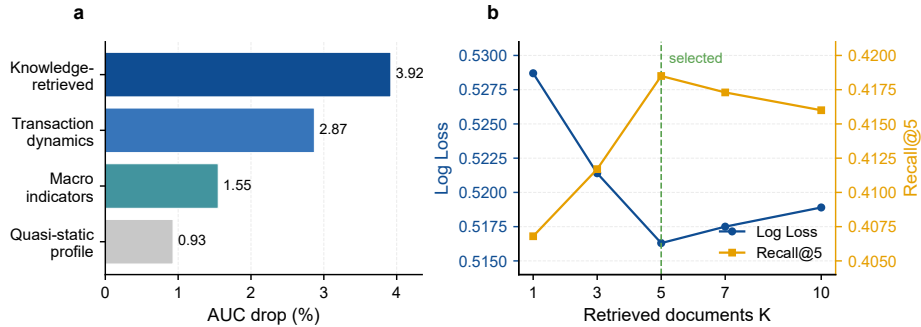


Fig. 3. Retrieval evidence analysis. (a) Feature-family occlusion measured by AUC drop. (b) Retrieval-depth sensitivity, with $K = 5$ selected for the main experiments.

4.6 Further Evaluation

Explanation Quality Beyond risk scores, FRAA is intended to support human review by explaining why a high-risk case was flagged. The goal is not to automate the final decision, but to make the model output easier to inspect. We therefore evaluate the generated explanations using both expert ratings and lexical overlap with reference explanations.

Five domain experts evaluated 1,000 high-risk test predictions. Figure 4 reports higher faithfulness, readability, and ROUGE-L for FRAA than for a template baseline that lists the top three feature groups. The full model obtains faithfulness of 4.32/5, readability of 4.51/5, and ROUGE-L of 0.4131. The result suggests that the explanation head adds information beyond feature enumeration. The expert ratings should be interpreted as an auditability signal, not as evidence that generated text is sufficient for automatic decision making.

Efficiency Analysis The final evaluation asks whether the added retrieval and fusion steps are practical for an analyst-facing review loop.

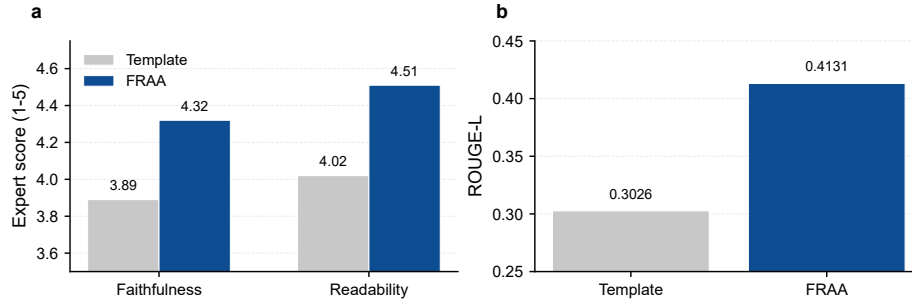


Fig. 4. Explanation-quality evaluation. (a) Expert-rated faithfulness and readability on a five-point scale. (b) ROUGE-L overlap with reference explanations.

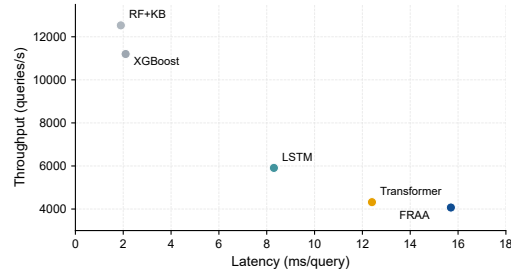


Fig. 5. Inference latency and throughput under the tested batch-size-64 serving configuration.

FRAA is slower than tree ensembles because it performs retrieval and neural sequence encoding. Even so, the tested latency in Fig. 5 remains compatible with interactive risk review: FRAA requires 15.7 ms per query and reaches 4,070 queries/s in the tested serving configuration. The retrieval step accounts for approximately 4.2 ms of the total latency. Throughput is lower than for tabular baselines, but remains above the level required for batched offline scoring and analyst triage in the tested configuration.

4.7 Limitations and Reproducibility

The evaluation uses a proprietary financial dataset that cannot be publicly released because it contains sensitive user-level risk and transaction information. This limits independent replication. To support reproducibility within this constraint, we report the data scale, feature families, chronological split, training configuration, retrieval depth, baselines, and ablation settings. The knowledge base is timestamped, and retrieval uses only documents available at or before the scoring time. These details make the experimental protocol auditable even when the raw records cannot be shared.

Several claims should be read within this scope. First, the results establish performance in one large multi-source financial environment, not universal transfer across institutions. Second, the current evaluation does not report multi-seed variance or formal statistical significance tests because the available audit record contains one fixed proprietary chronological split rather than repeated independent training runs. Third, the explanation evaluation measures expert-perceived faithfulness and readability for high-risk test predictions. It does not remove the need for human oversight in regulated decisions. Future work should add public benchmark evaluations where legally possible, repeatability analysis under multiple splits or random seeds, institution-level transfer tests including parameter-efficient adaptation methods such as LoRA [18], and governance mechanisms for retrieved evidence that may become outdated or disputed.

5 Conclusion

This paper presented FRAA, a retrieval-augmented model for explainable financial risk assessment. The main finding is that retrieval helps risk scoring most when external evidence is time-valid and fused with temporal user behavior, rather than attached as a static feature. On a large-scale proprietary multi-source dataset, FRAA obtains the lowest Log Loss and highest Recall@5 among the evaluated models. The ablation, occlusion, and retrieval-depth analyses trace the gains mainly to timestamp-valid external evidence, with five retrieved documents giving the best trade-off between evidence coverage and noise. Explanation and latency evaluations further indicate that the model can support analyst-facing review under the tested serving configuration. Taken together, the results suggest that external evidence retrieval is most valuable when it is constrained by time validity, fused with temporal user behavior, and exposed through auditable explanations.

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